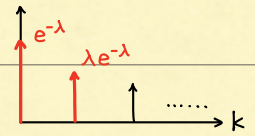
$X \sim Poi(\lambda)$ $\quad P(X=k) = \frac{1}{k!} e^{-\lambda} \cdot \lambda^k$

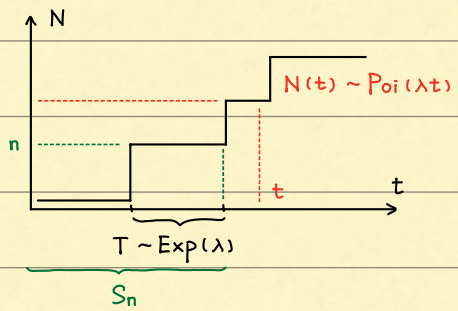


Poission Process Def

		Mean	Variance
$N(t) \sim Poi(\lambda t)$	$Poi(\lambda)$	λ	λ
$T \sim Exp(\lambda)$	$Exp(\lambda)$	$\frac{1}{\lambda}$	$(\frac{1}{\lambda})^2$

Increment stationary independent

$$P(N(t_2)=n_2 \mid N(t_1)=n_1) = P(N(\Delta t)=\Delta n)$$



$N(t) \sim Poi(\lambda t) \iff T \sim Exp(\lambda)$

Pf: $T \leq t \overset{\text{indp incr}}{\iff} S_n \leq t \iff N(t) \geq n$

发生 n 次时间 $\leq t \iff t$ 时刻发生次数 $\geq n$

$P(T \leq t) = 1 - P(N(t)=0) = 1 - e^{-\lambda t}$

$C(t_1, t_2) = \lambda \min(t_1, t_2)$

$E(N_t) = \lambda t$

$R(t_1, t_2) = E(N_{t_1} N_{t_2}) = E[N_{t_1}(N_{t_1} + N_\tau)]$ $\quad$ if $t_1 < t_2$

$= E(N_{t_1}^2) + E(N_{t_1})E(N_\tau) = \lambda t_1 + (\lambda t_1)^2 + \lambda t_1 \cdot \lambda(t_2 - t_1)$

$R(t_1, t_2) = \lambda \min(t_1, t_2) + \lambda^2 t_1 t_2$

$C(t_1, t_2) = R(t_1, t_2) - E(N_{t_1})E(N_{t_2})$

t 时发生 n 次，分配二项分布

$N_{t_1} \mid N_{t_2} = n \sim Binomial(n, \frac{t_1}{t_2})$ $\quad$ 与 λ 无关

$P(N_{t_1}=k \mid N_{t_2}=n) = P(N_{t_1}=k \cap N_\tau = n-k) / P(N_{t_2}=n)$

$$= \frac{1}{k!} e^{-\lambda t_1} \cdot (\lambda t_1)^k \cdot \frac{1}{(n-k)!} e^{-\lambda \tau} \cdot (\lambda \tau)^{n-k} / \frac{1}{n!} e^{-\lambda t_2} \cdot (\lambda t_2)^n$$

$$= \frac{n!}{k!(n-k)!} \frac{t_1^k \tau^{n-k}}{t_2^n} = \binom{n}{k} (\frac{t_1}{t_2})^k (\frac{\tau}{t_2})^{n-k}$$

e.g. Given $N(3)=5$，求 $t=1$ 时 2 次概率

$$P(N(1)=2 \mid N(3)=5) = \binom{5}{2}\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)^3$$

t 时发生一次，发生时间 $S_1 \sim U[0,t]$

e.g. Given $N(t)=n$, Find PDF of S_n

When $n=1$, $S_1 \sim U[0,t]$

CDF $F(s) = P(S_n \leq s \mid N(t)=n)$ $\quad S_n \leq s \iff N(s) \geq n$

$$P(N(s)=n \text{ and } N(t)=n) = P(N(s)=n)\,P(N(t-s)=0)$$

$$= \frac{1}{n!} e^{-\lambda s} (\lambda s)^n \cdot e^{-\lambda(t-s)}$$

$$P(N(t)=n) = \frac{1}{n!} e^{-\lambda t} (\lambda t)^n$$

$$F(s) = \frac{1}{n!} e^{-\lambda t} (\lambda s)^n \Big/ \frac{1}{n!} e^{-\lambda t} (\lambda t)^n = \left(\frac{s}{t}\right)^n$$

$$f(s) = F'(s) = \frac{n}{t^n} s^{n-1}, \quad 0 < s < t$$

When $n=1$, $f(s) = \frac{n}{t}$